Full rank of the gated MASaLA operator for generic parameters

Setting

Fix an input sequence $X \in \mathbb{R}^{N\times d}$ and collect all trainable parameters (projections, RACE projections, gate MLP, $\log\lambda$, layer norms, ...) in a vector $\theta \in \mathbb{R}^p$.

For a *support pattern* $\mathcal{S} = (S_1, \dots, S_N)$, where $S_i \subseteq \{1, \dots, N\}$ is the union of the key blocks selected for query i, define:

- the sparse attention matrix $P_S^{\mathcal{S}}(\theta)$ with $[P_S^{\mathcal{S}}]_{ij} = \exp(s_{ij})/d_{S,i}$ for $j \in S_i$ and 0 otherwise, where $s_{ij} = q_i^\top k_j/\sqrt{d_h}$ and $d_{S,i} = \sum_{j\in S_i}\exp(s_{ij})$;
- the RACE (low-rank) attention matrix $P_R(\theta)$, with $\operatorname{rank} P_R \le L_s 2^\gamma$, and its denominator $d_{R,i} > 0$;
- the mass multiplier $m_i = d_{S,i}/(d_{S,i} + \lambda d_{R,i} + \varepsilon)$;
- the gates $(g_{S,i}, g_{R,i}) = \sigma(G_\theta(x_i)) \in (0,1)^2$.

The gated (plain) operator used in the experiments is

$$A^{\mathcal{S}}(\theta) = \operatorname{diag}(g_S \odot m)\, P_S^{\mathcal{S}} + \operatorname{diag}(g_R)\, P_R,$$

and the ungated operator of Lemma 1 is $U^{\mathcal{S}}(\theta) = \operatorname{diag}(m)\, P_S^{\mathcal{S}} + P_R$. In the model the support is chosen by the block scorer, $\mathcal{S} = \mathcal{S}(\theta)$, and the operator actually applied is $A^{\mathcal{S}(\theta)}(\theta)$.

Assumptions

(A1) Structured support. Every admissible pattern contains each query's own block ($i \in S_i$), as in the paper (own block and first block are always selected).

(A2) Lemma 1. For every admissible pattern $\mathcal{S}$ there exists at least one θ_0 with $\det U^{\mathcal{S}}(\theta_0) \neq 0$.

(A3) Analyticity. Every map from θ to the quantities above is real-analytic on $\mathbb{R}^p$. This holds because they are compositions of linear maps, exp, tanh, the logistic sigmoid, SiLU, softmax normalisations with strictly positive denominators, $\lambda = \exp(\log\lambda)$, and LayerNorm (analytic because the variance is shifted by its $\epsilon > 0$). With QK-norm, $q/\|q\|$ is analytic wherever $q \neq 0$; restrict θ to that open set.

Theorem 1. *Under (A1)–(A3), the set $\mathcal{Z} = \{\theta \in \mathbb{R}^p : \det A^{\mathcal{S}(\theta)}(\theta) = 0\}$ has Lebesgue measure zero. In particular, if θ is drawn from any distribution with a density (for example a Gaussian initialisation), the gated attention matrix is full rank with probability one.*

Proof. **Step 1 (fixed pattern, analytic determinant).** Fix an admissible pattern $\mathcal{S}$. By (A3), every entry of $A^{\mathcal{S}}(\theta)$ is real-analytic in θ, so $f_{\mathcal{S}}(\theta) = \det A^{\mathcal{S}}(\theta)$ is real-analytic on $\mathbb{R}^p$, being a polynomial in the entries.

Step 2 (the determinant is not identically zero). Take θ_0 from (A2) and change only the last linear layer of the gate MLP: set its weights to zero and both biases to the same value b. The rest of the network is unchanged, so $P_S^{\mathcal{S}}$, P_R and m are the same as at θ_0, and every query gets $g_{S,i} = g_{R,i} = \sigma(b) =: c \in (0,1)$. Call this point θ_1. Then

$$A^{\mathcal{S}}(\theta_1) = c\big(\operatorname{diag}(m)P_S^{\mathcal{S}} + P_R\big) = c\,U^{\mathcal{S}}(\theta_0), \qquad f_{\mathcal{S}}(\theta_1) = c^N \det U^{\mathcal{S}}(\theta_0) \neq 0.$$

Step 3 (measure zero for one pattern). A real-analytic function on a connected open set that is not identically zero has a zero set of Lebesgue measure zero (Mityagin, *The zero set of a real analytic function*, Math. Notes 107, 2020). By Steps 1–2, $\mathcal{Z}_{\mathcal{S}} = \{\theta : f_{\mathcal{S}}(\theta) = 0\}$ has measure zero.

Step 4 (all patterns). The pattern actually used depends on θ through a top-k and is discontinuous, so we do not differentiate through it. We only use the inclusion

$$\mathcal{Z} \subseteq \bigcup_{\mathcal{S}\ \text{admissible}} \mathcal{Z}_{\mathcal{S}},$$

which holds because $\theta \in \mathcal{Z}$ implies $f_{\mathcal{S}(\theta)}(\theta) = 0$. With $n_b = N/B$ blocks and k selected blocks per query block there are at most $\binom{n_b}{k}^{n_b}$ patterns, a finite number. A finite union of measure-zero sets has measure zero. □

Remarks

- **Scope of "almost every".** The statement concerns generic parameters. A trained θ is a single point and is not random, so the theorem says that singular configurations are exceptional (measure zero, nowhere dense). It does not say that a particular trained model can never be singular.
- **Conditioning.** The result is about exact rank, not about the condition number. A gated matrix can be full rank but ill-conditioned, for example when $g_S \odot m$ is very small for some queries.
- **Arithmetic.** The argument is in exact arithmetic. In bf16 or fp32, "full rank" means non-singular up to numerical tolerance.
- **Other fusion modes.** The same argument applies to any fusion whose entries are analytic in θ and that reduces to a non-singular operator at one parameter value, for example the complement form, provided Lemma 1 (or an analogue) gives that witness.
- **Dependence on Lemma 1.** Step 2 needs only one non-singular witness per support pattern. Whatever assumptions Lemma 1 places on X (for example distinct tokens) carry over to this theorem.